

Vision

The written notes for the vision walkthrough. Week 5 competencies 16 to 19.

Focusing light

- Refraction is the bending of light. The cornea does about two thirds of it and has a fixed shape. The lens does the rest and can change shape.
- The lens is elastic and held by zonule fibers attached to the ring-shaped ciliary muscle.

	Far object	Near object (accommodation)
Ciliary muscle	Relaxed	Contracted
Zonules	Taut	Slack
Lens	Flat, less bending	Round, more bending

Pupillary light reflex

Light in one eye constricts both pupils. In: optic nerve. Center: midbrain. Out: parasympathetic fibers in the oculomotor nerve to the circular muscle of each iris. The response in the unlit eye is the consensual response.

Afferent defect

Light in the affected eye constricts neither pupil, light in the other eye constricts both: the problem is in the optic nerve of the affected eye.

Refractive errors

Condition	Cause	Focus falls	Blurred	Corrective lens
Myopia (nearsighted)	Eyeball too long or cornea too curved	In front of the retina	Distant objects	Concave
Hyperopia (farsighted)	Eyeball too short	Behind the retina	Near objects	Convex
Presbyopia	Lens stiffens with age, accommodation fails	Behind the retina for near objects	Near objects	Convex reading lens
Astigmatism	Cornea curved unevenly	Different for different directions	Lines in some directions	Cylindrical

Phototransduction

- Rhodopsin is opsin plus retinal (from vitamin A), in the discs of the rod outer segment.
- In the dark: cGMP is high, cGMP-gated cation channels are open, sodium enters, the rod sits near -40 mV and releases glutamate continuously.

1. A photon is absorbed and retinal changes shape.
2. Rhodopsin is activated.
3. Rhodopsin activates the G protein transducin.
4. Transducin activates phosphodiesterase.
5. Phosphodiesterase breaks down cGMP.
6. cGMP-gated channels close and sodium entry stops.
7. The photoreceptor hyperpolarizes toward -70 mV.
8. Fewer voltage-gated calcium channels at the base are open, so glutamate release falls.

- The cascade amplifies at each step, so a rod can respond to a single photon.
- Photoreceptors make graded potentials only. The first action potentials are in the ganglion cells, whose axons form the optic nerve.

Rods, cones and the visual pathway

	Rods	Cones
Light needed	Dim (very sensitive)	Bright
Color	No	Yes
Where	Most of the retina outside the fovea	Packed in the fovea
Convergence onto ganglion cells	High: sensitive, low acuity	Low, often one to one: high acuity

1. Optic nerve from each eye.
2. Optic chiasm: fibers from the inner (nasal) half of each retina cross; fibers from the outer half stay on their own side.
3. Optic tract to the thalamus (lateral geniculate nucleus).
4. Visual cortex in the occipital lobe.

Lesion	Field loss
One optic nerve, before the chiasm	All vision in that eye
Middle of the chiasm (pituitary tumor)	Outer (temporal) half of each eye's field: bitemporal hemianopia, often unnoticed for a long time